

DATE:18.08.26

EX:NO:16

Decision Tree Graph Output

AIM

To implement a Decision Tree and represent the decision-making process graphically.

ALGORITHM

1. Create a simple decision tree.
2. Define the root, decision and leaf nodes.
3. Connect the nodes.
4. Display the tree using Matplotlib.

CODE

```
import matplotlib.pyplot as plt
```

```
# Node positions
```

```
pos = {
```

```
    "Weather": (0, 4),
```

```
    "Sunny": (-2, 3),
```

```
    "Rainy": (2, 3),
```

```
    "Hot": (-3, 2),
```

```
    "Cold": (-1, 2),
```

```
    "Play": (2, 2),
```

```
    "No": (-3, 1),
```

```
    "Yes": (-1, 1)
```

```
}
```

```
# Connections
```

```
edges = [  
    ("Weather", "Sunny"),  
    ("Weather", "Rainy"),  
    ("Sunny", "Hot"),  
    ("Sunny", "Cold"),  
    ("Hot", "No"),  
    ("Cold", "Yes"),  
    ("Rainy", "Play"),  
    ("Play", "Yes")  
]
```

```
plt.figure(figsize=(9, 6))
```

```
# Draw edges
```

```
for start, end in edges:
```

```
    x1, y1 = pos[start]
```

```
    x2, y2 = pos[end]
```

```
    plt.plot([x1, x2], [y1, y2], 'k-')
```

```
# Draw nodes
```

```
for node, (x, y) in pos.items():
```

```
if node in ["Yes", "No"]:
    size = 1400
else:
    size = 1800

plt.scatter(x, y, s=size)
plt.text(
    x, y, node,
    ha="center",
    va="center",
    fontsize=11
)

# Edge labels
plt.text(-1.2, 3.5, "Sunny", fontsize=10)
plt.text(1.0, 3.5, "Rainy", fontsize=10)
plt.text(-2.7, 2.5, "Hot", fontsize=10)
plt.text(-1.4, 2.5, "Cold", fontsize=10)

plt.title("Decision Tree")
plt.axis("off")
plt.show()

print("Decision Tree successfully implemented.")
```

nds + Code + Text ▶ Run all

